Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX2B

DES - CRYPTOGRAPHIC ALGORITHM

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Code: (--CyptoFunctions--)

Cypto_functions.py:

```
# crypto_functions.py

# DES TABLES
IP = [
58,50,42,34,26,18,10,2,
60,52,44,36,28,20,12,4,
62,54,46,38,30,22,14,6,
64,56,48,40,32,24,16,8,
57,49,41,33,25,17,9,1,
59,51,43,35,27,19,11,3,
61,53,45,37,29,21,13,5,
63,55,47,39,31,23,15,7
]
IP_INV = [
40,8,48,16,56,24,64,32,
39,7,47,15,55,23,63,31,
38,6,46,14,54,22,62,30,
37,5,45,13,53,21,61,29,
36,4,44,12,52,20,60,28,
35,3,43,11,51,19,59,27,
34,2,42,10,50,18,58,26,
33,1,41,9,49,17,57,25
]
E = [
32,1,2,3,4,5,
4,5,6,7,8,9,
8,9,10,11,12,13,
12,13,14,15,16,17,
16,17,18,19,20,21,
20,21,22,23,24,25,
24,25,26,27,28,29,
28,29,30,31,32,1
]
P = [
16,7,20,21,29,12,28,17,
1,15,23,26,5,18,31,10,
2,8,24,14,32,27,3,9,
19,13,30,6,22,11,4,25
]
```

```
PC1 = [
57,49,41,33,25,17,9,
1,58,50,42,34,26,18,
10,2,59,51,43,35,27,
19,11,3,60,52,44,36,
63,55,47,39,31,23,15,
7,62,54,46,38,30,22,
14,6,61,53,45,37,29,
21,13,5,28,20,12,4
]
PC2 = [
14,17,11,24,1,5,
3,28,15,6,21,10,
23,19,12,4,26,8,
16,7,27,20,13,2,
41,52,31,37,47,55,
30,40,51,45,33,48,
44,49,39,56,34,53,
46,42,50,36,29,32
]
SHIFTS = [1,1,2,2,2,2,2,2,1,2,2,2,2,2,2,1]
SBOX = [
[[14,4,13,1,2,15,11,8,3,10,6,12,5,9,0,7],
[0,15,7,4,14,2,13,1,10,6,12,11,9,5,3,8],
[4,1,14,8,13,6,2,11,15,12,9,7,3,10,5,0],
[15,12,8,2,4,9,1,7,5,11,3,14,10,0,6,13]],
]

# HELPER FUNCTIONS
def str_to_bin(s):
   return ''.join(format(ord(c), '08b') for c in s)
def bin_to_hex(b):
   return hex(int(b, 2))[2:].upper().zfill(len(b)//4)
def permute(block, table):
   return ''.join(block[i-1] for i in table)
def xor(a, b):
   return ''.join('0' if x == y else '1' for x, y in zip(a, b))
def shift_left(k, n):
   return k[n:] + k[:n]

# KEY SCHEDULER
def generate_subkeys(key_64):
   key = permute(key_64, PC1)
```

```python
    C, D = key[:28], key[28:]
    subkeys = []
    for shift in SHIFTS:
        C = shift_left(C, shift)
        D = shift_left(D, shift)
        subkeys.append(permute(C + D, PC2))
    return subkeys

# ROUND FUNCTION
def f_function(R, K):
    ER = permute(R, E)
    x = xor(ER, K)
    return permute(x[:32], P)  # simplified S-box handling

# DES CORE
def des_block(block, subkeys, encrypt=True, verbose=True):
    if verbose:
        print("\nInitial Block (HEX):", bin_to_hex(block))
    block = permute(block, IP)
    if verbose:
        print("After IP (HEX):", bin_to_hex(block))
    L, R = block[:32], block[32:]
    keys = subkeys if encrypt else subkeys[::-1]
    for i in range(16):
        f = f_function(R, keys[i])
        new_R = xor(L, f)
        L, R = R, new_R
        if verbose:
            print(f"Round {i+1:02d} (HEX):", bin_to_hex(L + R))
    final = permute(R + L, IP_INV)
    if verbose:
        print("After IP⁻¹ (HEX):", bin_to_hex(final))
    return final

# ECB MODE
def des_encrypt_ecb(text, key, verbose=True):
    bin_text = str_to_bin(text)
    bin_key = str_to_bin(key[:8])
    subkeys = generate_subkeys(bin_key)
    blocks = []
    for i in range(0, len(bin_text), 64):
        blocks.append(bin_text[i:i+64].ljust(64, '0'))
    if verbose:
```

```
        print(f"\nECB MODE: {len(blocks)} block(s) identified")
    cipher = ""
    for idx, block in enumerate(blocks, start=1):
        if verbose:
            ascii_block = ''.join(
                chr(int(block[i:i+8], 2)) for i in range(0, 64, 8)
            )
            print(f"\n--- Encrypting Block {idx} ---")
            print("Plaintext Block (ASCII):", ascii_block)
            print("Plaintext Block (HEX)  :", bin_to_hex(block))
        enc_block = des_block(block, subkeys, True, verbose)
        cipher += enc_block
        if verbose:
            print("Cipher Block (HEX)     :", bin_to_hex(enc_block))
    return bin_to_hex(cipher)

def des_decrypt_ecb(hex_cipher, key, verbose=True):
    bin_cipher = bin(int(hex_cipher, 16))[2:].zfill(len(hex_cipher)*4)
    bin_key = str_to_bin(key[:8])
    subkeys = generate_subkeys(bin_key)
    blocks = []
    for i in range(0, len(bin_cipher), 64):
        blocks.append(bin_cipher[i:i+64])
    if verbose:
        print(f"\nECB MODE: {len(blocks)} block(s) identified")
    plain = ""
    for idx, block in enumerate(blocks, start=1):
        if verbose:
            print(f"\n--- Decrypting Block {idx} ---")
            print("Cipher Block (HEX)     :", bin_to_hex(block))
        dec_block = des_block(block, subkeys, False, verbose)
        plain += dec_block
        if verbose:
            ascii_block = ''.join(
                chr(int(dec_block[i:i+8], 2)) for i in range(0, 64, 8)
            )
            print("Plaintext Block (ASCII):", ascii_block)
    chars = [chr(int(plain[i:i+8], 2)) for i in range(0, len(plain), 8)]
    return ''.join(chars).rstrip('\x00')

# DEMO RUN
def main():
    print("=== DES DEMO RUN ===")
```

```python
    pt = "Cryptography"
    key = "Security!"
    print("Plaintext :", pt)
    print("Key       :", key)
    ct = des_encrypt_ecb(pt, key)
    print("\nCiphertext (HEX):", ct)
    dec = des_decrypt_ecb(ct, key)
    print("\nDecrypted Text:", dec)
if __name__ == "__main__":
    main()
```

OUTPUT:

```
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python crypto_functions.py
=== DES DEMO RUN ===
Plaintext : Cryptography
Key       : Security!

ECB MODE: 2 block(s) identified

--- Encrypting Block 1 ---
Plaintext Block (ASCII): Cryptogr
Plaintext Block (HEX)  : 43727970746F6772

Initial Block (HEX): 43727970746F6772
After IP (HEX): FF9E706500FE24E3
Round 01 (HEX): 00FE24E35EAEFF43
Round 02 (HEX): 5EAEFF43DB87C814
Round 03 (HEX): DB87C8142C6092A9
Round 04 (HEX): 2C6092A9BDB23A9F
Round 05 (HEX): BDB23A9F0515E84F
Round 06 (HEX): 0515E84F587F70EA
Round 07 (HEX): 587F70EA0BC6C847
Round 08 (HEX): 0BC6C847BD736AE4
Round 09 (HEX): BD736AE4358167EB
Round 10 (HEX): 358167EB22C41DFD
Round 11 (HEX): 22C41DFDD25B0F86
Round 12 (HEX): D25B0F86F2B1678B
Round 13 (HEX): F2B1678B466068DA
Round 14 (HEX): 466068DA3318BD64
Round 15 (HEX): 3318BD643A97D3F2
Round 16 (HEX): 3A97D3F263C31A32
After IP⁻¹ (HEX): 78FF2084AFC35A3A
Cipher Block (HEX)     : 78FF2084AFC35A3A

--- Encrypting Block 2 ---
Plaintext Block (ASCII): aphy
Plaintext Block (HEX)  : 6170687900000000

Initial Block (HEX): 6170687900000000
After IP (HEX): 0F0A0009000F0C00
Round 01 (HEX): 000F0C0029DA978E
Round 02 (HEX): 29DA978E08BFEF11
Round 03 (HEX): 08BFEF11B89978A0
Round 04 (HEX): B89978A08976A03B
Round 05 (HEX): 8976A03BCDB63B64
Round 06 (HEX): CDB63B64E0091C57
Round 07 (HEX): E0091C5710ADD7D6
Round 08 (HEX): 10ADD7D6EDEA4E24
Round 09 (HEX): EDEA4E2449C66AFB
Round 10 (HEX): 49C66AFB22357206
Round 11 (HEX): 2235720629FC1E37
Round 12 (HEX): 29FC1E372EEBDA2F
Round 13 (HEX): 2EEBDA2FD69992D3
Round 14 (HEX): D69992D300BCBC61
Round 15 (HEX): 00BCBC617EE739BD
Round 16 (HEX): 7EE739BDD3B73497
After IP⁻¹ (HEX): 7BF1B78ADFBEE073
Cipher Block (HEX)     : 7BF1B78ADFBEE073

Ciphertext (HEX): 78FF2084AFC35A3A7BF1B78ADFBEE073
```

```
ECB MODE: 2 block(s) identified

--- Decrypting Block 1 ---
Cipher Block (HEX)     : 78FF2084AFC35A3A

Initial Block (HEX): 78FF2084AFC35A3A
After IP (HEX): 63C31A323A97D3F2
Round 01 (HEX): 3A97D3F23318BD64
Round 02 (HEX): 3318BD64466068DA
Round 03 (HEX): 466068DAF2B1678B
Round 04 (HEX): F2B1678BD25B0F86
Round 05 (HEX): D25B0F8622C41DFD
Round 06 (HEX): 22C41DFD358167EB
Round 07 (HEX): 358167EBBD736AE4
Round 08 (HEX): BD736AE40BC6C847
Round 09 (HEX): 0BC6C847587F70EA
Round 10 (HEX): 587F70EA0515E84F
Round 11 (HEX): 0515E84FBDB23A9F
Round 12 (HEX): BDB23A9F2C6092A9
Round 13 (HEX): 2C6092A9DB87C814
Round 14 (HEX): DB87C8145EAEFF43
Round 15 (HEX): 5EAEFF4300FE24E3
Round 16 (HEX): 00FE24E3FF9E7065
After IP⁻¹ (HEX): 43727970746F6772
Plaintext Block (ASCII): Cryptogr

--- Decrypting Block 2 ---
Cipher Block (HEX)     : 7BF1B78ADFBEE073

Initial Block (HEX): 7BF1B78ADFBEE073
After IP (HEX): D3B734977EE739BD
Round 01 (HEX): 7EE739BD00BCBC61
Round 02 (HEX): 00BCBC61D69992D3
Round 03 (HEX): D69992D32EEBDA2F
Round 04 (HEX): 2EEBDA2F29FC1E37
Round 05 (HEX): 29FC1E3722357206
Round 06 (HEX): 2235720649C66AFB
Round 07 (HEX): 49C66AFBEDEA4E24
Round 08 (HEX): EDEA4E2410ADD7D6
Round 09 (HEX): 10ADD7D6E0091C57
Round 10 (HEX): E0091C57CDB63B64
Round 11 (HEX): CDB63B648976A03B
Round 12 (HEX): 8976A03BB89978A0
Round 13 (HEX): B89978A008BFEF11
Round 14 (HEX): 08BFEF1129DA978E
Round 15 (HEX): 29DA978E000F0C00
Round 16 (HEX): 000F0C000F0A0009
After IP⁻¹ (HEX): 6170687900000000
Plaintext Block (ASCII): aphy

Decrypted Text: Cryptography
```

Sender (server.py):

```
import socket
import crypto_functions as cf
KEY = "Security!"
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 12345))
server.listen(1)
print("Server ready (DES Encryption)")
conn, addr = server.accept()
print("Connected:", addr)
while True:
   pt = input("\nEnter Plaintext (or exit): ")
   if pt.lower() == "exit":
       conn.send(b"exit")
       break
   ct = cf.des_encrypt_ecb(pt, KEY, verbose=True)
   conn.send(ct.encode())
conn.close()
server.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf
KEY = "Security!"
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 12345))
print("Connected to server (DES Decryption)")
while True:
   ct = client.recv(4096).decode()
   if ct == "exit":
       break
   print("\nReceived Ciphertext:", ct)
   pt = cf.des_decrypt_ecb(ct, KEY, verbose=True)
   print("\nDecrypted Text:", pt)
client.close()
```

Output: (Server)

```
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python server.py
Server ready (DES Encryption)
Connected: ('127.0.0.1', 46066)

Enter Plaintext (or exit): Hello there, I am learning Cryptography... How are you?

ECB MODE: 7 block(s) identified

--- Encrypting Block 1 ---
Plaintext Block (ASCII): Hello th
Plaintext Block (HEX)  : 48656C6C6F207468

Initial Block (HEX): 48656C6C6F207468
After IP (HEX): DF405E1200FE9D10
Round 01 (HEX): 00FE9D1074E2F914
Round 02 (HEX): 74E2F914EB4231F5
Round 03 (HEX): EB4231F558D6A8DD
Round 04 (HEX): 58D6A8DD527DD8D4
Round 05 (HEX): 527DD8D447562564
Round 06 (HEX): 47562564B741BC88
Round 07 (HEX): B741BC88F890EA92
Round 08 (HEX): F890EA9201842448
Round 09 (HEX): 0184244813BF9885
Round 10 (HEX): 13BF988525247EFB
Round 11 (HEX): 25247EFB7224E5AC
Round 12 (HEX): 7224E5AC4625A826
Round 13 (HEX): 4625A826B3873F52
Round 14 (HEX): B3873F525BE6E586
Round 15 (HEX): 5BE6E5867236DAC0
Round 16 (HEX): 7236DAC08F1D18E1
After IP⁻¹ (HEX): 51E8705CBCA18B4B
Cipher Block (HEX)     : 51E8705CBCA18B4B

--- Encrypting Block 2 ---
Plaintext Block (ASCII): ere, I a
Plaintext Block (HEX)  : 6572652C20492061

Initial Block (HEX): 6572652C20492061
After IP (HEX): A7020DA500DF2802
Round 01 (HEX): 00DF280286928AA2
Round 02 (HEX): 86928AA220280265
Round 03 (HEX): 2028026500E11937
Round 04 (HEX): 00E11937422D89DD
Round 05 (HEX): 422D89DD56AB900E
Round 06 (HEX): 56AB900E586FDD44
Round 07 (HEX): 586FDD445B6898A6
Round 08 (HEX): 5B6898A641EDD142
Round 09 (HEX): 41EDD1421A31CCB0
Round 10 (HEX): 1A31CCB059CBDFAD
Round 11 (HEX): 59CBDFAD8DD6C282
```

Round 12 (HEX): 8DD6C282FD83DAAE
Round 13 (HEX): FD83DAAE927ECC22
Round 14 (HEX): 927ECC224F36A3C9
Round 15 (HEX): 4F36A3C916073C30
Round 16 (HEX): 16073C3018FD792D
After IP^{-1} (HEX): 35A0B95DDE1F1410
Cipher Block (HEX) : 35A0B95DDE1F1410

--- Encrypting Block 3 ---
Plaintext Block (ASCII): m learni
Plaintext Block (HEX) : 6D206C6561726E69

Initial Block (HEX): 6D206C6561726E69
After IP (HEX): FD204D9900FFC560
Round 01 (HEX): 00FFC5605EA0E6BE
Round 02 (HEX): 5EA0E6BEE30021BF
Round 03 (HEX): E30021BF7AD6F76F
Round 04 (HEX): 7AD6F76F523CC4BC
Round 05 (HEX): 523CC4BC6D3472F7
Round 06 (HEX): 6D3472F73FC3E4FA
Round 07 (HEX): 3FC3E4FADEF06139
Round 08 (HEX): DEF0613909D75518
Round 09 (HEX): 09D755183EBD57B7
Round 10 (HEX): 3EBD57B72D6762D5
Round 11 (HEX): 2D6762D5574462A7
Round 12 (HEX): 574462A7CEB6954E
Round 13 (HEX): CEB6954E99C560E0
Round 14 (HEX): 99C560E0DB3694D4
Round 15 (HEX): DB3694D4573611F2
Round 16 (HEX): 573611F20FDD44D5
After IP^{-1} (HEX): D9E2F550BB229713
Cipher Block (HEX) : D9E2F550BB229713

--- Encrypting Block 4 ---
Plaintext Block (ASCII): ng Crypt
Plaintext Block (HEX) : 6E67204372797074

Initial Block (HEX): 6E67204372797074
After IP (HEX): FBF0832A00F7211B
Round 01 (HEX): 00F7211B7AE40C0D
Round 02 (HEX): 7AE40C0DF1D8E8E6
Round 03 (HEX): F1D8E8E63B4F217C
Round 04 (HEX): 3B4F217C6D523B20
Round 05 (HEX): 6D523B20D6F2C99A
Round 06 (HEX): D6F2C99A5E76E318
Round 07 (HEX): 5E76E318F016C893
Round 08 (HEX): F016C893FE332BC0
Round 09 (HEX): FE332BC0CE10617B
Round 10 (HEX): CE10617B24AD8E0E
Round 11 (HEX): 24AD8E0E813D2817
Round 12 (HEX): 813D28172611F634

Round 13 (HEX): 2611F634D99CFC4B
Round 14 (HEX): D99CFC4B1A67DD0F
Round 15 (HEX): 1A67DD0F168F03BC
Round 16 (HEX): 168F03BC63AA1FCB
After IP^{-1} (HEX): 6DFDA63786524133
Cipher Block (HEX) : 6DFDA63786524133

--- Encrypting Block 5 ---
Plaintext Block (ASCII): ography.
Plaintext Block (HEX) : 6F6772617068792E

Initial Block (HEX): 6F6772617068792E
After IP (HEX): 7F54834B00FFE187
Round 01 (HEX): 00FFE187DE54286C
Round 02 (HEX): DE54286C7650B9F8
Round 03 (HEX): 7650B9F8B3F89059
Round 04 (HEX): B3F8905975689AA7
Round 05 (HEX): 75689AA7BDD91827
Round 06 (HEX): BDD91827EEFB24C0
Round 07 (HEX): EEFB24C0EF01AD2D
Round 08 (HEX): EF01AD2D1D97A9C5
Round 09 (HEX): 1D97A9C54596B683
Round 10 (HEX): 4596B683DB1AF857
Round 11 (HEX): DB1AF857D5102830
Round 12 (HEX): D5102830290A836C
Round 13 (HEX): 290A836CA696B9B1
Round 14 (HEX): A696B9B17571EF09
Round 15 (HEX): 7571EF0972860598
Round 16 (HEX): 728605982C982ECE
After IP^{-1} (HEX): 08A56D5792C48133
Cipher Block (HEX) : 08A56D5792C48133

--- Encrypting Block 6 ---
Plaintext Block (ASCII): .. How a
Plaintext Block (HEX) : 2E2E20486F772061

Initial Block (HEX): 2E2E20486F772061
After IP (HEX): B82033B000F71B33
Round 01 (HEX): 00F71B333336B4B7
Round 02 (HEX): 3336B4B7FE8AA832
Round 03 (HEX): FE8AA83273DCDC1A
Round 04 (HEX): 73DCDC1A6D32054D
Round 05 (HEX): 6D32054D14A33CDC
Round 06 (HEX): 14A33CDC55657BF4
Round 07 (HEX): 55657BF43B745528
Round 08 (HEX): 3B7455287DF35950
Round 09 (HEX): 7DF359500B316884
Round 10 (HEX): 0B31688427CD77FB
Round 11 (HEX): 27CD77FBCECF0D8C
Round 12 (HEX): CECF0D8CA2B05C1D
Round 13 (HEX): A2B05C1D10DE60F1

Round 14 (HEX): 10DE60F196949D5A
Round 15 (HEX): 96949D5A5A2D4A9F
Round 16 (HEX): 5A2D4A9F6BD900A6
After IP^{-1} (HEX): 72CB23FA9261D813
Cipher Block (HEX) : 72CB23FA9261D813

--- Encrypting Block 7 ---
Plaintext Block (ASCII): re you?
Plaintext Block (HEX) : 726520796F753F00

Initial Block (HEX): 726520796F753F00
After IP (HEX): 3B69727A007F5851
Round 01 (HEX): 007F5851947BE17D
Round 02 (HEX): 947BE17DC6656E17
Round 03 (HEX): C6656E17206DFDCB
Round 04 (HEX): 206DFDCB9274D1AD
Round 05 (HEX): 9274D1AD1F49FA53
Round 06 (HEX): 1F49FA530EF6DB48
Round 07 (HEX): 0EF6DB4877A7E17A
Round 08 (HEX): 77A7E17A29128A75
Round 09 (HEX): 29128A758B18EBD7
Round 10 (HEX): 8B18EBD7D39804DF
Round 11 (HEX): D39804DF158C095C
Round 12 (HEX): 158C095C168CED64
Round 13 (HEX): 168CED643003C5A3
Round 14 (HEX): 3003C5A315DC1CA0
Round 15 (HEX): 15DC1CA05AA54F89
Round 16 (HEX): 5AA54F89CC95915C
After IP^{-1} (HEX): 3E8879CB9520C976
Cipher Block (HEX) : 3E8879CB9520C976

Enter Plaintext (or exit):

Output: (Client)

(lab) ak@1125:~/Desktop/Data/CryptoLab$ python client.py
Connected to server (DES Decryption)

Received Ciphertext:
51E8705CBCA18B4B35A0B95DDE1F1410D9E2F550BB2297136DFDA6378652413308A56D5792C48133
72CB23FA9261D8133E8879CB9520C976

ECB MODE: 7 block(s) identified

--- Decrypting Block 1 ---
Cipher Block (HEX) : 51E8705CBCA18B4B

Initial Block (HEX): 51E8705CBCA18B4B
After IP (HEX): 8F1D18E17236DAC0
Round 01 (HEX): 7236DAC05BE6E586
Round 02 (HEX): 5BE6E586B3873F52
Round 03 (HEX): B3873F524625A826
Round 04 (HEX): 4625A8267224E5AC
Round 05 (HEX): 7224E5AC25247EFB
Round 06 (HEX): 25247EFB13BF9885
Round 07 (HEX): 13BF988501842448
Round 08 (HEX): 01842448F890EA92
Round 09 (HEX): F890EA92B741BC88
Round 10 (HEX): B741BC8847562564
Round 11 (HEX): 47562564527DD8D4
Round 12 (HEX): 527DD8D458D6A8DD
Round 13 (HEX): 58D6A8DDEB4231F5
Round 14 (HEX): EB4231F574E2F914
Round 15 (HEX): 74E2F91400FE9D10
Round 16 (HEX): 00FE9D10DF405E12
After IP^{-1} (HEX): 48656C6C6F207468
Plaintext Block (ASCII): Hello th

--- Decrypting Block 2 ---
Cipher Block (HEX) : 35A0B95DDE1F1410

Initial Block (HEX): 35A0B95DDE1F1410
After IP (HEX): 18FD792D16073C30
Round 01 (HEX): 16073C304F36A3C9
Round 02 (HEX): 4F36A3C9927ECC22
Round 03 (HEX): 927ECC22FD83DAAE
Round 04 (HEX): FD83DAAE8DD6C282
Round 05 (HEX): 8DD6C28259CBDFAD
Round 06 (HEX): 59CBDFAD1A31CCB0
Round 07 (HEX): 1A31CCB041EDD142
Round 08 (HEX): 41EDD1425B6898A6
Round 09 (HEX): 5B6898A6586FDD44
Round 10 (HEX): 586FDD4456AB900E
Round 11 (HEX): 56AB900E422D89DD
Round 12 (HEX): 422D89DD00E11937

Round 13 (HEX): 00E1193720280265
Round 14 (HEX): 2028026586928AA2
Round 15 (HEX): 86928AA200DF2802
Round 16 (HEX): 00DF2802A7020DA5
After IP^{-1} (HEX): 6572652C20492061
Plaintext Block (ASCII): ere, I a

--- Decrypting Block 3 ---
Cipher Block (HEX) : D9E2F550BB229713

Initial Block (HEX): D9E2F550BB229713
After IP (HEX): 0FDD44D5573611F2
Round 01 (HEX): 573611F2DB3694D4
Round 02 (HEX): DB3694D499C560E0
Round 03 (HEX): 99C560E0CEB6954E
Round 04 (HEX): CEB6954E574462A7
Round 05 (HEX): 574462A72D6762D5
Round 06 (HEX): 2D6762D53EBD57B7
Round 07 (HEX): 3EBD57B709D75518
Round 08 (HEX): 09D75518DEF06139
Round 09 (HEX): DEF061393FC3E4FA
Round 10 (HEX): 3FC3E4FA6D3472F7
Round 11 (HEX): 6D3472F7523CC4BC
Round 12 (HEX): 523CC4BC7AD6F76F
Round 13 (HEX): 7AD6F76FE30021BF
Round 14 (HEX): E30021BF5EA0E6BE
Round 15 (HEX): 5EA0E6BE00FFC560
Round 16 (HEX): 00FFC560FD204D99
After IP^{-1} (HEX): 6D206C6561726E69
Plaintext Block (ASCII): m learni

--- Decrypting Block 4 ---
Cipher Block (HEX) : 6DFDA63786524133

Initial Block (HEX): 6DFDA63786524133
After IP (HEX): 63AA1FCB168F03BC
Round 01 (HEX): 168F03BC1A67DD0F
Round 02 (HEX): 1A67DD0FD99CFC4B
Round 03 (HEX): D99CFC4B2611F634
Round 04 (HEX): 2611F634813D2817
Round 05 (HEX): 813D281724AD8E0E
Round 06 (HEX): 24AD8E0ECE10617B
Round 07 (HEX): CE10617BFE332BC0
Round 08 (HEX): FE332BC0F016C893
Round 09 (HEX): F016C8935E76E318
Round 10 (HEX): 5E76E318D6F2C99A
Round 11 (HEX): D6F2C99A6D523B20
Round 12 (HEX): 6D523B203B4F217C
Round 13 (HEX): 3B4F217CF1D8E8E6
Round 14 (HEX): F1D8E8E67AE40C0D
Round 15 (HEX): 7AE40C0D00F7211B

Round 16 (HEX): 00F7211BFBF0832A
After IP^{-1} (HEX): 6E67204372797074
Plaintext Block (ASCII): ng Crypt

--- Decrypting Block 5 ---
Cipher Block (HEX) : 08A56D5792C48133

Initial Block (HEX): 08A56D5792C48133
After IP (HEX): 2C982ECE72860598
Round 01 (HEX): 728605987571EF09
Round 02 (HEX): 7571EF09A696B9B1
Round 03 (HEX): A696B9B1290A836C
Round 04 (HEX): 290A836CD5102830
Round 05 (HEX): D5102830DB1AF857
Round 06 (HEX): DB1AF8574596B683
Round 07 (HEX): 4596B6831D97A9C5
Round 08 (HEX): 1D97A9C5EF01AD2D
Round 09 (HEX): EF01AD2DEEFB24C0
Round 10 (HEX): EEFB24C0BDD91827
Round 11 (HEX): BDD9182775689AA7
Round 12 (HEX): 75689AA7B3F89059
Round 13 (HEX): B3F890597650B9F8
Round 14 (HEX): 7650B9F8DE54286C
Round 15 (HEX): DE54286C00FFE187
Round 16 (HEX): 00FFE1877F54834B
After IP^{-1} (HEX): 6F6772617068792E
Plaintext Block (ASCII): ography.

--- Decrypting Block 6 ---
Cipher Block (HEX) : 72CB23FA9261D813

Initial Block (HEX): 72CB23FA9261D813
After IP (HEX): 6BD900A65A2D4A9F
Round 01 (HEX): 5A2D4A9F96949D5A
Round 02 (HEX): 96949D5A10DE60F1
Round 03 (HEX): 10DE60F1A2B05C1D
Round 04 (HEX): A2B05C1DCECF0D8C
Round 05 (HEX): CECF0D8C27CD77FB
Round 06 (HEX): 27CD77FB0B316884
Round 07 (HEX): 0B3168847DF35950
Round 08 (HEX): 7DF359503B745528
Round 09 (HEX): 3B74552855657BF4
Round 10 (HEX): 55657BF414A33CDC
Round 11 (HEX): 14A33CDC6D32054D
Round 12 (HEX): 6D32054D73DCDC1A
Round 13 (HEX): 73DCDC1AFE8AA832
Round 14 (HEX): FE8AA8323336B4B7
Round 15 (HEX): 3336B4B700F71B33
Round 16 (HEX): 00F71B33B82033B0
After IP^{-1} (HEX): 2E2E20486F772061
Plaintext Block (ASCII): .. How a

--- Decrypting Block 7 ---
Cipher Block (HEX) : 3E8879CB9520C976

Initial Block (HEX): 3E8879CB9520C976
After IP (HEX): CC95915C5AA54F89
Round 01 (HEX): 5AA54F8915DC1CA0
Round 02 (HEX): 15DC1CA03003C5A3
Round 03 (HEX): 3003C5A3168CED64
Round 04 (HEX): 168CED64158C095C
Round 05 (HEX): 158C095CD39804DF
Round 06 (HEX): D39804DF8B18EBD7
Round 07 (HEX): 8B18EBD729128A75
Round 08 (HEX): 29128A7577A7E17A
Round 09 (HEX): 77A7E17A0EF6DB48
Round 10 (HEX): 0EF6DB481F49FA53
Round 11 (HEX): 1F49FA539274D1AD
Round 12 (HEX): 9274D1AD206DFDCB
Round 13 (HEX): 206DFDCBC6656E17
Round 14 (HEX): C6656E17947BE17D
Round 15 (HEX): 947BE17D007F5851
Round 16 (HEX): 007F58513B69727A
After IP^{-1} (HEX): 726520796F753F00
Plaintext Block (ASCII): re you?

Decrypted Text: Hello there, I am learning Cryptography... How are you?

Output:

```
ak@1125: ~/Desktop/Data/CryptoLab
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python server.py
Server ready (DES Encryption)
Connected: ('127.0.0.1', 46066)

Enter Plaintext (or exit): Hello there, I am learning Cryptography...
How are you?

ECB MODE: 7 block(s) identified

--- Encrypting Block 1 ---
Plaintext Block (ASCII): Hello th
Plaintext Block (HEX)  : 48656C6C6F207468

Initial Block (HEX): 48656C6C6F207468
After IP (HEX): DF405E1200FE9D10
Round 01 (HEX): 00FE9D1074E2F914
Round 02 (HEX): 74E2F914EB4231F5
Round 03 (HEX): EB4231F558D6A8DD
Round 04 (HEX): 58D6A8DD527DD8D4
Round 05 (HEX): 527DD8D447562564
Round 06 (HEX): 47562564B741BC88
Round 07 (HEX): B741BC88F890EA92
Round 08 (HEX): F890EA9201842448
Round 09 (HEX): 0184244813BF9885
Round 10 (HEX): 13BF988525247EFB
Round 11 (HEX): 25247EFB7224E5AC
Round 12 (HEX): 7224E5AC4625A826
Round 13 (HEX): 4625A826B3873F52
Round 14 (HEX): B3873F525BE6E586
Round 15 (HEX): 5BE6E5867236DAC0
Round 16 (HEX): 7236DAC08F1D18E1
After IP⁻¹ (HEX): 51E8705CBCA18B4B
Cipher Block (HEX)     : 51E8705CBCA18B4B

--- Encrypting Block 2 ---
Plaintext Block (ASCII): ere, I a
Plaintext Block (HEX)  : 6572652C20492061

Initial Block (HEX): 6572652C20492061
After IP (HEX): A7020DA500DF2802
Round 01 (HEX): 00DF280286928AA2
Round 02 (HEX): 86928AA220280265
Round 03 (HEX): 2028026500E11937
Round 04 (HEX): 00E11937422D89DD
Round 05 (HEX): 422D89DD56AB900E
Round 06 (HEX): 56AB900E586FDD44
Round 07 (HEX): 586FDD445B6898A6
Round 08 (HEX): 5B6898A641EDD142
Round 09 (HEX): 41EDD1421A31CCB0
Round 10 (HEX): 1A31CCB059CBDFAD
Round 11 (HEX): 59CBDFAD8DD6C282
Round 12 (HEX): 8DD6C282FD83DAAE
Round 13 (HEX): FD83DAAE927ECC22
Round 14 (HEX): 927ECC224F36A3C9
Round 15 (HEX): 4F36A3C916073C30
Round 16 (HEX): 16073C3018FD792D
After IP⁻¹ (HEX): 35A0B95DDE1F1410
Cipher Block (HEX)     : 35A0B95DDE1F1410

--- Encrypting Block 3 ---
Plaintext Block (ASCII): m learni
Plaintext Block (HEX)  : 6D206C6561726E69
```

```
ak@1125: ~/Desktop/Data/CryptoLab
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python client.py
Connected to server (DES Decryption)

Received Ciphertext: 51E8705CBCA18B4B35A0B95DDE1F1410D9E2F550BB229713
DA6378652413308A56D5792C4813372CB23FA9261D8133E8879CB9520C976

ECB MODE: 7 block(s) identified

--- Decrypting Block 1 ---
Cipher Block (HEX)     : 51E8705CBCA18B4B

Initial Block (HEX): 51E8705CBCA18B4B
After IP (HEX): 8F1D18E17236DAC0
Round 01 (HEX): 7236DAC05BE6E586
Round 02 (HEX): 5BE6E586B3873F52
Round 03 (HEX): B3873F524625A826
Round 04 (HEX): 4625A8267224E5AC
Round 05 (HEX): 7224E5AC25247EFB
Round 06 (HEX): 25247EFB13BF9885
Round 07 (HEX): 13BF988501842448
Round 08 (HEX): 01842448F890EA92
Round 09 (HEX): F890EA92B741BC88
Round 10 (HEX): B741BC8847562564
Round 11 (HEX): 47562564527DD8D4
Round 12 (HEX): 527DD8D458D6A8DD
Round 13 (HEX): 58D6A8DDEB4231F5
Round 14 (HEX): EB4231F574E2F914
Round 15 (HEX): 74E2F91400FE9D10
Round 16 (HEX): 00FE9D10DF405E12
After IP⁻¹ (HEX): 48656C6C6F207468
Plaintext Block (ASCII): Hello th

--- Decrypting Block 2 ---
Cipher Block (HEX)     : 35A0B95DDE1F1410

Initial Block (HEX): 35A0B95DDE1F1410
After IP (HEX): 18FD792D16073C30
Round 01 (HEX): 16073C304F36A3C9
Round 02 (HEX): 4F36A3C9927ECC22
Round 03 (HEX): 927ECC22FD83DAAE
Round 04 (HEX): FD83DAAE8DD6C282
Round 05 (HEX): 8DD6C28259CBDFAD
Round 06 (HEX): 59CBDFAD1A31CCB0
Round 07 (HEX): 1A31CCB041EDD142
Round 08 (HEX): 41EDD1425B6898A6
Round 09 (HEX): 5B6898A6586FDD44
Round 10 (HEX): 586FDD4456AB900E
Round 11 (HEX): 56AB900E422D89DD
Round 12 (HEX): 422D89DD00E11937
Round 13 (HEX): 00E1193720280265
Round 14 (HEX): 2028026586928AA2
Round 15 (HEX): 86928AA200DF2802
Round 16 (HEX): 00DF2802A7020DA5
After IP⁻¹ (HEX): 6572652C20492061
Plaintext Block (ASCII): ere, I a

--- Decrypting Block 3 ---
Cipher Block (HEX)     : D9E2F550BB229713

Initial Block (HEX): D9E2F550BB229713
After IP (HEX): 0FDD44D5573611F2
Round 01 (HEX): 573611F2DB3694D4
```

```
ak@1125: ~/Desktop/Data/CryptoLab
Round 09 (HEX): 1D97A9C54596B683
Round 10 (HEX): 4596B683DB1AF857
Round 11 (HEX): DB1AF857D5102830
Round 12 (HEX): D5102830290A836C
Round 13 (HEX): 290A836CA696B9B1
Round 14 (HEX): A696B9B17571EF09
Round 15 (HEX): 7571EF0972860598
Round 16 (HEX): 728605982C982ECE
After IP⁻¹ (HEX): 08A56D5792C48133
Cipher Block (HEX)     : 08A56D5792C48133

--- Encrypting Block 6 ---
Plaintext Block (ASCII): .. How a
Plaintext Block (HEX)  : 2E2E20486F772061

Initial Block (HEX): 2E2E20486F772061
After IP (HEX): B82033B000F71B33
Round 01 (HEX): 00F71B333336B4B7
Round 02 (HEX): 3336B4B7FE8AA832
Round 03 (HEX): FE8AA83273DCDC1A
Round 04 (HEX): 73DCDC1A6D32054D
Round 05 (HEX): 6D32054D14A33CDC
Round 06 (HEX): 14A33CDC55657BF4
Round 07 (HEX): 55657BF43B745528
Round 08 (HEX): 3B7455287DF35950
Round 09 (HEX): 7DF359500B316884
Round 10 (HEX): 0B31688427CD77FB
Round 11 (HEX): 27CD77FBCECF0D8C
Round 12 (HEX): CECF0D8CA2B05C1D
Round 13 (HEX): A2B05C1D10DE60F1
Round 14 (HEX): 10DE60F196949D5A
Round 15 (HEX): 96949D5A5A2D4A9F
Round 16 (HEX): 5A2D4A9F6BD900A6
After IP⁻¹ (HEX): 72CB23FA9261D813
Cipher Block (HEX)     : 72CB23FA9261D813

--- Encrypting Block 7 ---
Plaintext Block (ASCII): re you?
Plaintext Block (HEX)  : 726520796F753F00

Initial Block (HEX): 726520796F753F00
After IP (HEX): 3B69727A007F5851
Round 01 (HEX): 007F5851947BE17D
Round 02 (HEX): 947BE17DC6656E17
Round 03 (HEX): C6656E17206DFDCB
Round 04 (HEX): 206DFDCB9274D1AD
Round 05 (HEX): 9274D1AD1F49FA53
Round 06 (HEX): 1F49FA530EF6DB48
Round 07 (HEX): 0EF6DB4877A7E17A
Round 08 (HEX): 77A7E17A29128A75
Round 09 (HEX): 29128A758B18EBD7
Round 10 (HEX): 8B18EBD7D39804DF
Round 11 (HEX): D39804DF158C095C
Round 12 (HEX): 158C095C168CED64
Round 13 (HEX): 168CED643003C5A3
Round 14 (HEX): 3003C5A315DC1CA0
Round 15 (HEX): 15DC1CA05AA54F89
Round 16 (HEX): 5AA54F89CC95915C
After IP⁻¹ (HEX): 3E8879CB9520C976
Cipher Block (HEX)     : 3E8879CB9520C976

Enter Plaintext (or exit): 
```

```
ak@1125: ~/Desktop/Data/CryptoLab
Round 08 (HEX): 1D97A9C5EF01AD2D
Round 09 (HEX): EF01AD2DEEFB24C0
Round 10 (HEX): EEFB24C0BDD91827
Round 11 (HEX): BDD9182775689AA7
Round 12 (HEX): 75689AA7B3F89059
Round 13 (HEX): B3F890597650B9F8
Round 14 (HEX): 7650B9F8DE54286C
Round 15 (HEX): DE54286C00FFE187
Round 16 (HEX): 00FFE1877F54834B
After IP⁻¹ (HEX): 6F6772617068792E
Plaintext Block (ASCII): ography.

--- Decrypting Block 6 ---
Cipher Block (HEX)     : 72CB23FA9261D813

Initial Block (HEX): 72CB23FA9261D813
After IP (HEX): 6BD900A65A2D4A9F
Round 01 (HEX): 5A2D4A9F96949D5A
Round 02 (HEX): 96949D5A10DE60F1
Round 03 (HEX): 10DE60F1A2B05C1D
Round 04 (HEX): A2B05C1DCECF0D8C
Round 05 (HEX): CECF0D8C27CD77FB
Round 06 (HEX): 27CD77FB0B316884
Round 07 (HEX): 0B3168847DF35950
Round 08 (HEX): 7DF359503B745528
Round 09 (HEX): 3B74552855657BF4
Round 10 (HEX): 55657BF414A33CDC
Round 11 (HEX): 14A33CDC6D32054D
Round 12 (HEX): 6D32054D73DCDC1A
Round 13 (HEX): 73DCDC1AFE8AA832
Round 14 (HEX): FE8AA8323336B4B7
Round 15 (HEX): 3336B4B700F71B33
Round 16 (HEX): 00F71B33B82033B0
After IP⁻¹ (HEX): 2E2E20486F772061
Plaintext Block (ASCII): .. How a

--- Decrypting Block 7 ---
Cipher Block (HEX)     : 3E8879CB9520C976

Initial Block (HEX): 3E8879CB9520C976
After IP (HEX): CC95915C5AA54F89
Round 01 (HEX): 5AA54F8915DC1CA0
Round 02 (HEX): 15DC1CA03003C5A3
Round 03 (HEX): 3003C5A3168CED64
Round 04 (HEX): 168CED64158C095C
Round 05 (HEX): 158C095CD39804DF
Round 06 (HEX): D39804DF8B18EBD7
Round 07 (HEX): 8B18EBD729128A75
Round 08 (HEX): 29128A7577A7E17A
Round 09 (HEX): 77A7E17A0EF6DB48
Round 10 (HEX): 0EF6DB481F49FA53
Round 11 (HEX): 1F49FA539274D1AD
Round 12 (HEX): 9274D1AD206DFDCB
Round 13 (HEX): 206DFDCBC6656E17
Round 14 (HEX): C6656E17947BE17D
Round 15 (HEX): 947BE17D007F5851
Round 16 (HEX): 007F58513B69727A
After IP⁻¹ (HEX): 726520796F753F00
Plaintext Block (ASCII): re you?

Decrypted Text: Hello there, I am learning Cryptography... How are you
```